

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2273

Jefferson Street 138 kV
November 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2273. The Interconnection Customer proposes a 359.4 MW battery energy storage facility interconnecting at the Jefferson Street 138 kV substation in the Virginia area. The request was received on January 05, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 759 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$27,950,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHAPMAN SUBSTATION to CARNESTOWN 138kV Line 1	CHAPMAN to CARNESTOWN 138kV Circuit 2	127.2	271	41.4
MULLIN to COSBY 138kV L	COSBY to MUSTANG BAYOU 138kV	125.7	244	5.5
DELTA PUMPS to CHAPMAN 138kV #1	DELTA PUMPS to CHAPMAN 138kV #2	120.9	257	40.6
CHAPMAN to DELTA PUMPS 138kV Cir. 2	DELTA PUMPS to CHAPMAN 138kV Line 1	120.9	257	40.6
CHAPMAN to CARNESTOWN 138kV	CHAPMAN to CARNESTOWN 138kV	117.0	294	41.4
MUSTANG BAYOU to COSBY 138kV L	COSBY to MULLIN 138kV Line	113.9	241	5.4
FALU 1334 to BURROWS CROSSING ROAD 138kV L	FALU 1334 to BURROWS CROSSING ROAD 138kV Circuit 1	113.2	497	8.3
JEFFERSON ST to DELTA PUMPS 138kV Cir. 1	JEFFERSON ST to DELTA PUMPS 138kV Line 2	112.2	255	40.8
DELTA PUMPS to JEFFERSON STREET 138kV	JEFFERSON STREET SUBSTATION to DELTA PUMPS SUBSTATION 138kV Cir. 1	112.2	255	40.8
COSBY to FIVE FORKS 138kV L	ROBERTS SW TA 345 kV/1 kV XFMR	108.0	270	12.7
FALU 1334 to BURROWS CROSSING RD SUBSTATION 138kV #1	FALU 1334 to BURROWS CROSSING RD 138kV line	105.0	536	8.3
HAYFIELD to BELVOIR 138kV Line 1	HAYFIELD to BELVOIR 138kV	103.9	255	5.1
HAYFIELD to BELVOR 138kV Circuit 2	BELVOIR to HAYFIELD 138kV Circuit 1	103.9	255	5.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PATHFINDER SUBSTATION to CARREON 138kV Line	BROOKRIDGE to BURROWS CROSSING ROAD 138kV line	103.3	240	6.9
VAN DOM to HAYFIELD 138kV	BURNT CHURCH to FALU 1334 138kV	102.9	336	7.7
NEWTON to NEW HOPE 138kV	CENTENNIAL SUBSTATION to ANNANDALE 138kV	102.1	215	12.7
PATHFINDER SUBSTATION to DOS PALOS 138kV line	BROOKRIDGE to BURROWS CROSSING ROAD SUBSTATION 138kV	101.3	238	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHAPMAN to CARNESTOWN 138kV	Rebuild line with higher-capacity conductor	\$1,574,000	95% (\$1,500,000)
MULLIN SUBSTATION to COSBY 138kV	Rebuild line with higher-capacity conductor	\$10,307,000	100% (\$10,307,000)
CHAPMAN to DELTA PUMPS 138kV Line 1	Rebuild line with higher-capacity conductor	\$861,000	95% (\$820,000)
CHAPMAN to DELTA PUMPS 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$751,000	95% (\$715,000)
CHAPMAN to CARNESTOWN 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,279,000	95% (\$1,218,000)
MUSTANG BAYOU to COSBY 138kV	Rebuild line with higher-capacity conductor	\$5,020,000	100% (\$5,020,000)
FALU 1334 to BURROWS CROSSING ROAD 138kV L	Rebuild line with higher-capacity conductor	\$1,161,000	28% (\$319,000)
DELT PUMPS to JEFFERSON STREET 138kV Line 1	Rebuild line with higher-capacity conductor	\$1,388,000	95% (\$1,322,000)
DELTA PUMPS to JEFFERSON STREET 138kV	Rebuild line with higher-capacity conductor	\$989,000	95% (\$942,000)
COSBY to FIVE FORKS SUBSTATION 138kV L	Reconductor line	\$2,282,000	100% (\$2,282,000)
FALU 1334 to BURROWS CROSSING RD 138kV Cir. 1	Reconductor line	\$538,000	28% (\$148,000)
HAYFIELD to BELVOIR 138kV Line 1	Reconductor line	\$550,000	100% (\$550,000)
BELVOIR to HAYFIELD 138kV Cir. 2	Reconductor line	\$727,000	100% (\$727,000)
PATHFINDER SUBSTATION to CARREON 138kV Line	Reconductor line	\$1,599,000	64% (\$1,027,000)
VAN DOM to HAYFIELD 138kV	Reconductor line	\$545,000	26% (\$141,000)
NEW HOPE to NEWTON 138kV line	Reconductor line	\$611,000	60% (\$370,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PATHFINDER SUBSTATION to DOS PALOS 138kV line	Reconductor line	\$844,000	64% (\$542,000)
	Total allocated to project		\$27,950,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.